

AYESH ABU LEHYEH

Ph.D. Candidate in Computer Science · Computer Vision and Visual Geo-Localization

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SUMMARY

Computer vision researcher working on visual geo-localization: locating ground-level images by matching them to satellite imagery, from cross-view retrieval to real-time, fine-grained pose estimation. I also work on localization from RF signals with calibrated uncertainty and on vision-language reasoning for spatial understanding. First-author papers at CVPR, IEEE IoT Journal, IEEE ICC, and a NeurIPS 2026 workshop.

Research interests: cross-view image geo-localization, 3D vision (reconstruction, feature matching, pose estimation), generative models (flow matching, diffusion), vision-language models, multimodal localization.

EDUCATION

University of Vermont

Ph.D. in Computer Science, Advisor: Prof. Safwan Wshah

Sep 2023 – Expected Aug 2027

Burlington, VT

University of Vermont

M.S. in Computer Science, GPA: 4.0/4.0

2023 – 2025

Burlington, VT

- Coursework: Machine Learning, Deep Learning, Data Mining, Data Science, Algorithm Design, LLMs.

University of Jordan

B.S. in Electrical Engineering

2014 – 2019

Amman, Jordan

RESEARCH EXPERIENCE

University of Vermont

Graduate Research Assistant

Sep 2023 – Present

Burlington, VT

Cross-view image geo-localization

- **GeoFlow (CVPR 2026):** reframed fine-grained ground-to-satellite localization as iterative, flow-inspired refinement, where many pose hypotheses converge to a consensus, with an inference scheme that trades accuracy for speed at test time without retraining. Delivered real-time fine-grained localization with an efficient architecture.([page](#)).
- **GeoSync (under review, ICLR 2027):** showed that cross-view retrieval and fine-grained localization can be learned from one shared representation instead of a two-model cascade, with localization evidence re-ranking retrieval and built-in confidence signals for failure-aware geo-localization. Achieved state-of-the-art end-to-end accuracy on VIGOR and DReSS at lower inference cost than cascaded pipelines.
- **Vision-language reasoning (NeurIPS 2026 workshop):** studied what spatial information survives when images become language, using a training-free pipeline in which a vision-language model describes ground and satellite views as structured text for retrieval, LLM judging, and reranking. Found that scene structure transfers into language while the fine detail needed to separate nearby places does not ([code](#)).
- **Weather robustness:** Benchmarked state-of-the-art cross-view retrieval under severe weather and seasonal shifts, and proposed a robust model for these conditions. ([code](#)).

RF and wireless localization

- **Conformal indoor localization (IEEE ICC 2026):** made conformal prediction spatially adaptive inside a graph transformer, so uncertainty regions adjust to local Wi-Fi signal conditions instead of using one global margin. Produced location-aware uncertainty with statistical coverage guarantees.
- **UBiGTLoc (IEEE IoT Journal 2025):** unified anchor-free and anchor-based sensor localization in one framework by modeling how RSSI changes over time (BiLSTM) and how nodes relate in space (GNN). Outperformed existing methods in dense and sparse networks using only low-cost RSSI, without GPS-equipped anchors.

Industry project with GlobalFoundries

- Framed chip quality prediction from thousands of RF and DC tests across production lots as a learning problem, showed the test space is highly redundant, and built models that predict RF pass/fail from cheap DC tests, opening a path to cutting costly high-frequency RF testing in semiconductor manufacturing.

INDUSTRY EXPERIENCE

NextNav

AI/ML Intern, Positioning, Navigation, and Timing

Jun 2026 – Aug 2026

Remote, USA

- Replaced the fixed lookup table that sets measurement trust in 5G positioning with probabilistic ML models that learn a full error distribution for each measurement from raw channel responses, so the position solver can down-weight multipath-corrupted signals.
- Trained these models end-to-end through a differentiable position solver, so learning targets final position accuracy rather than a proxy loss, cutting tail position error by up to **25%** in field tests.

Orange Jordan

Senior Data Analyst (2021 – 2023); RF Engineer (2019 – 2021)

May 2019 – Aug 2023

Amman, Jordan

- **Senior Data Analyst:** built machine learning models for anomaly detection and performance analysis on large-scale network data, with Python and SQL pipelines, to surface network degradations automatically.
- **RF Engineer:** optimized radio network coverage, performance, and reliability across 3G, 4G, and 5G networks.

PUBLICATIONS

Conference and journal papers

- [1] **Ayesh Abu Lehyeh**, Xiaohan Zhang, Ahmad Arrabi, Waqas Sultani, Chen Chen, Safwan Wshah. **GeoFlow: Real-Time Fine-Grained Cross-View Geolocalization via Iterative Flow Prediction**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [[project](#)] [[arXiv](#)]
- [2] **Ayesh Abu Lehyeh**, Anastassia Gharib, Safwan Wshah. **Spatially-Adaptive Conformal Graph Transformer for Indoor Localization in Wi-Fi Driven Networks**. *IEEE International Conference on Communications (ICC)*, 2026. [[paper](#)] [[arXiv](#)]
- [3] **Ayesh Abu Lehyeh**, Anastassia Gharib, Tian Xia, Dryver Huston, Safwan Wshah. **UBiGTLoc: A Unified BiLSTM-Graph Transformer Localization Framework for IoT Sensor Networks**. *IEEE Internet of Things Journal*, 2025. [[paper](#)] [[arXiv](#)]
- [4] Hakan Unveren, **Ayesh Abu Lehyeh**, Duygu Pamukcu, Christopher W. Zobel. **Strategies for Crisis-Responsive Governance: Automated Anomaly Identification in Public Services**. *International ISCRAM Conference*, 2024. [[paper](#)]

Workshop papers

- [5] **Ayesh Abu Lehyeh**, Jay Hwasung Jung, Safwan Wshah. **What Words Keep of a Place: Zero-Shot Language Reasoning for Cross-View Geo-Localization**. *NeurIPS 2026 Workshop on Physical World AI*. [[arXiv](#)] [[code](#)]

Under review

- [6] **Ayesh Abu Lehyeh**, et al. **GeoSync: Unified Representation Learning for End-to-End Cross-View Geolocalization**. Under review at ICLR 2027.

TECHNICAL SKILLS

Computer Vision	Cross-view geo-localization, image retrieval, fine-grained localization, pose estimation, BEV
Machine Learning	Representation learning, transformers, graph neural networks, vision-language models, probabilistic modeling, uncertainty quantification, conformal prediction
Wireless and PNT	5G and RF positioning, Wi-Fi localization, robust trilateration, radio network optimization
Programming	Python, PyTorch, PyTorch Geometric, C++, SQL, MATLAB, Linux, Git, Slurm/HPCC

ACADEMIC SERVICE

Reviewer	NeurIPS 2026, ICLR 2027, IEEE TPAMI, IEEE TNSE, ECAI 2025, ISPRS Journal
Sub-reviewer	CVPR 2025, CVPR 2026, ECCV 2026, AAAI 2027, TGRS Journal